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Student Name: .....

Enrolment No. 15611622

Mid-Term Examination – Nov 2023

Programme: B.Tech (AIML/AI&DS)

Paper Code: AIML/AI&DS-209

Time: One and Half Hours

Semester: Third Semester

Paper Name: Probability, Statistics & Linear Algebra

Maximum Marks: 30

Note:

Question no. 1 is compulsory

Attempt any two questions from the remaining questions

All questions carry equal marks

Only scientific calculator is allowed

	Question 1	
1 (a)	Define probability space with example.	2
1 (b)	A die is thrown twice and the sum of the number appearing is observed to be 6. What is the probability that the number 4 has appeared at least once.	2
1 (c)	Write a short note on skewness of a distribution.	2
1 (d)	X is a Poisson variable and it is found that the probability at X=2 is 2/3 of the probability at X=1. Find the probability at X=3.	2
1 (e)	Find the moment generating function (m.g.f. about origin) of the exponential distribution $f(x) = \frac{1}{c}e^{x/c}$, $0 \le x \le \infty, c>0$.	2

	Question 2	
2 (a)	A factory has 2 plants. Records shows that the plan 1 produces 30% and plant 2 produces 70% items. Further 5% of the items produced by plant 1 are defective while 1% produced by plant 2 are defective. If a defective item is drawn at random, find the probability that the defective item was produced by: (i) Plant 1 (ii) Plant 2	5
2 (b)	A random variable X has the following probability mass function: $p(x) = \begin{cases} x/15 & x = 1,2,3,4,5 \\ 0 & otherwise \end{cases}$ Find: (i) P{X=1 or 2} (ii) P{1/2 < X < 5/2 | X > 1}	5

	Question 3	
3 (a)	Fit a binomial distribution for the following data and compare the theoretical frequencies with actual once.	5

x	0	1	2	3	4	5
f	2	14	20	34	22	8

3 (b)	For a distribution, the mean is 10, variance is 16, γ_1 is 1 and β_2 is 4. Obtain the first four moments about origin. Comment upon the nature of distribution.	5

	Question 4	
4 (a)	If the heights of 300 students are normally distributed with mean 68 inch and standard deviation 3 inch. How many students have height: (i) Greater than 72 inches (ii) Between 65 and 71 inches Given P(z≤1.33=0.4082), P(z≤1=0.3413), P(z≥-1=0.3413)	5
4 (b)	Obtain the rank correlation coefficient between the variables X and Y from the following pairs of observed data.	5

X	50	55	65	50	55	60	50	65	70	75
Y	110	110	115	125	140	115	130	120	115	160